

1 a.e.

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. We define

$$L^1(\Omega) = \left\{ f : \Omega \rightarrow \mathbb{R} : \int_{\Omega} |f| \, d\mu < \infty \right\}.$$

Then $f \mapsto \int_{\Omega} f \, d\mu$ is a linear functional on $L^1(\Omega)$.

Remark 1.1. If f is integrable, then

$$\mu(\{\omega \in \Omega : |f(\omega)| = \infty\}) = 0.$$

Thus an integrable function is finite *almost everywhere*.

Proof. Let

$$E := \{\omega \in \Omega : |f(\omega)| = \infty\}.$$

Then, for every $k > 0$,

$$E \subseteq \{\omega \in \Omega : |f(\omega)| > k\}.$$

Hence $|f(\omega)| \geq k \mathbf{1}_E(\omega)$, and therefore

$$\int_{\Omega} |f| \, d\mu \geq k\mu(E) \implies \mu(E) \leq \frac{1}{k} \int_{\Omega} |f| \, d\mu.$$

As $k \rightarrow \infty$, we have $\mu(E) = 0$ □

2 Fatou's Lemma

Theorem 2.1 (Fatou's Lemma). Let $f_n : \Omega \rightarrow [0, \infty]$ be measurable and $f_n \geq 0$. Then

$$\int_{\Omega} \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu.$$

Note. Observe that Fatou's Lemma requires very few conditions to apply (*non-negativity* and *measurability* of f_n) compared to other powerful theorems. In practise, this means that spotting when the lemma is useful very difficult.

Proof. Define $g_n := \inf_{k \geq n} f_k$. Then g_n is measurable and $g_n \uparrow \liminf_{n \rightarrow \infty} f_n$. By the Monotone Convergence Theorem,

$$\int_{\Omega} \liminf_{n \rightarrow \infty} f_n \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu.$$

Now, for every $k \geq n$, $g_n \leq f_k$ so,

$$\int_{\Omega} g_n \, d\mu \leq \inf_{k \geq n} \int_{\Omega} f_k \, d\mu.$$

Therefore

$$\begin{aligned} \int_{\Omega} \liminf_{n \rightarrow \infty} f_n \, d\mu &= \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu \\ &\leq \lim_{n \rightarrow \infty} \inf_{k \geq n} \int_{\Omega} f_k \, d\mu \\ &= \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \end{aligned}$$

□

3 Dominated Convergence Theorem

Theorem 3.1 (Dominated Convergence Theorem). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f_n : \Omega \rightarrow \mathbb{R}$ be measurable. Suppose f_n converges pointwise, i.e.

$$f_n(\omega) \rightarrow f(\omega) \quad \forall \omega \in \Omega,$$

and there exists a measurable function $g : \Omega \rightarrow [0, \infty]$ such that

$$|f_n(\omega)| \leq g(\omega) \quad \forall \omega \in \Omega, \quad \text{with} \quad \int_{\Omega} g \, d\mu < \infty.$$

Then f is integrable and

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu = \int_{\Omega} f \, d\mu.$$

and in particular,

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f_n - f| \, d\mu = 0.$$

Proof. Since g is integrable, f_n and f are integrable. Hence $f_n - f$ is integrable and

$$\left| \int_{\Omega} f_n \, d\mu - \int_{\Omega} f \, d\mu \right| = \left| \int_{\Omega} (f_n - f) \, d\mu \right| \leq \int_{\Omega} |f_n - f| \, d\mu.$$

Also we have the bound, $|f_n - f| \leq |f_n| + |f| \leq 2g$. We define,

$$h_n(\omega) := 2g(\omega) - |f_n(\omega) - f(\omega)|.$$

Then $h_n \geq 0$ and h_n is measurable. Moreover, since $f_n \rightarrow f$, $h_n(\omega) \rightarrow 2g(\omega)$. Applying Fatou's Lemma to (h_n) , we obtain

$$\begin{aligned} 2 \int_{\Omega} g \, d\mu &= \int_{\Omega} \lim_{n \rightarrow \infty} h_n \, d\mu \\ &\leq \liminf_{n \rightarrow \infty} \int_{\Omega} h_n \, d\mu \\ &= \liminf_{n \rightarrow \infty} \left(2 \int_{\Omega} g \, d\mu - \int_{\Omega} |f_n - f| \, d\mu \right) \\ &= 2 \int_{\Omega} g \, d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} |f_n - f| \, d\mu. \end{aligned}$$

Therefore

$$\limsup_{n \rightarrow \infty} \int_{\Omega} |f_n - f| \, d\mu \leq 0.$$

Since the integrals are non-negative,

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f_n - f| \, d\mu = 0. \implies \boxed{\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu = \int_{\Omega} f \, d\mu.}$$

□

Example 3.2. Consider $((0, 1), \mathcal{B}_{(0,1)}, \lambda)$ and define $f_n = n \mathbf{1}_{(0, 1/n]}$. Then

$$f_n(x) \rightarrow 0 =: f(x) \quad \forall x \in (0, 1),$$

and

$$\int_{(0,1)} f_n \, d\lambda = n\lambda((0, 1/n]) = 1,$$

whereas $\int_{(0,1)} f \, d\lambda = 0$. Thus pointwise convergence alone does not imply

$$\int f_n \, d\mu \rightarrow \int f \, d\mu.$$

Example 3.3. Consider $(\mathbb{N}, \mathcal{P}(\mathbb{N}), \#)$ where $\#$ denotes counting measure. Let $f_n = \mathbf{1}_{[n, \infty)}$. Then $f_n(k) \rightarrow 0 \ \forall k \in \mathbb{N}$ but

$$\int_{\mathbb{N}} f_n \, d\# = \infty \quad \forall n,$$

while

$$\int_{\mathbb{N}} 0 \, d\# = 0.$$

Theorem 3.4. Let $E \in \mathcal{F}$ with $\mu(E^c) = 0$. Then, for every integrable f ,

$$\int_{\Omega} f \, d\mu = \int_E f \, d\mu.$$

Proof. Since E^c is a null set, $\int_{E^c} f \, d\mu = 0$. Hence

$$\int_{\Omega} f \, d\mu = \int_E f \, d\mu + \int_{E^c} f \, d\mu = \int_E f \, d\mu.$$

□

Theorem 3.5 (Lebesgue's MCT). Suppose $f_n \geq 0$ a.s. and $f_n \uparrow f$ a.s. Then

$$\int_{\Omega} \lim_{n \rightarrow \infty} f_n \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu.$$

Corollary 3.6. If $\mu(\{\omega \in \Omega : f(\omega) \neq g(\omega)\}) = 0$ then

$$\int_{\Omega} f \, d\mu = \int_{\Omega} g \, d\mu.$$

Definition 3.7. Let $f : \Omega \rightarrow \overline{\mathbb{R}}$. We say that f is *almost surely measurable* if there exists a measurable function $g : \Omega \rightarrow \overline{\mathbb{R}}$ such that

$$\mu(\{\omega \in \Omega : f(\omega) \neq g(\omega)\}) = 0.$$

Theorem 3.8. Suppose $f_n : \Omega_n \rightarrow \overline{\mathbb{R}}$ with $\mu(\Omega_n^c) = 0 \ \forall n \geq 1$ and f_n is almost surely measurable. Assume

$$\sum_{n=1}^{\infty} \int_{\Omega} |f_n| \, d\mu < \infty.$$

Then $\sum_{n=1}^{\infty} f_n(\omega)$ converges absolutely almost surely. Moreover,

$$\sum_{n=1}^{\infty} \int_{\Omega} f_n \, d\mu = \int_{\Omega} \sum_{n=1}^{\infty} f_n \, d\mu.$$

Proof. Find $E \in \mathcal{F}$ such that $\mu(E^c) = 0$ and $E \subseteq \Omega_n \ \forall n \geq 1$. For each n , choose a measurable function g_n which agrees with f_n on E .

Define

$$g(\omega) := \left(\sum_{k=1}^{\infty} |f_k(\omega)| \right) \mathbf{1}_E(\omega), \quad \text{and} \quad g_n(\omega) := \left(\sum_{k=1}^n |f_k(\omega)| \right) \mathbf{1}_E(\omega) \quad \text{for each } n$$

Then g_n is measurable and $g_n \uparrow g$. By the MCT,

$$\int_{\Omega} g \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu.$$

Moreover,

$$\begin{aligned}
\int_{\Omega} g \, d\mu &= \lim_{n \rightarrow \infty} \int_{\Omega} \left(\sum_{k=1}^n |f_k| \right) \mathbf{1}_E \, d\mu \\
&= \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_{\Omega} |f_k| \mathbf{1}_E \, d\mu \\
&= \sum_{k=1}^{\infty} \int_{\Omega} |f_k| \, d\mu < \infty.
\end{aligned}$$

Hence $g(\omega) < \infty$ a.e. Therefore $\sum_{n=1}^{\infty} |f_n(\omega)| < \infty$ a.e. and thus $\sum_{n=1}^{\infty} f_n(\omega)$ converges absolutely a.e. Now put $h_n = \sum_{k=1}^n f_k \mathbf{1}_E$, then

$$\begin{aligned}
h_n &\rightarrow \left(\sum_{k=1}^{\infty} f_k \right) \mathbf{1}_E \quad \text{a.e.} \\
\implies |h_n| &\leq \sum_{k=1}^{\infty} |f_k| \mathbf{1}_E = g
\end{aligned}$$

where $g \in L^1(\Omega)$. By the DCT,

$$\lim_{n \rightarrow \infty} \int_{\Omega} h_n \, d\mu = \int_{\Omega} \left(\sum_{k=1}^{\infty} f_k \right) \mathbf{1}_E \, d\mu.$$

Since $\mu(E^c) = 0$,

$$\int_{\Omega} \left(\sum_{k=1}^{\infty} f_k \right) \mathbf{1}_E \, d\mu = \int_{\Omega} \sum_{k=1}^{\infty} f_k \, d\mu.$$

On the other hand,

$$\int_{\Omega} h_n \, d\mu = \sum_{k=1}^n \int_{\Omega} f_k \, d\mu. \implies \boxed{\sum_{n=1}^{\infty} \int_{\Omega} f_n \, d\mu = \int_{\Omega} \sum_{n=1}^{\infty} f_n \, d\mu.}$$

□